

ENGINEERING PORTFOLIO

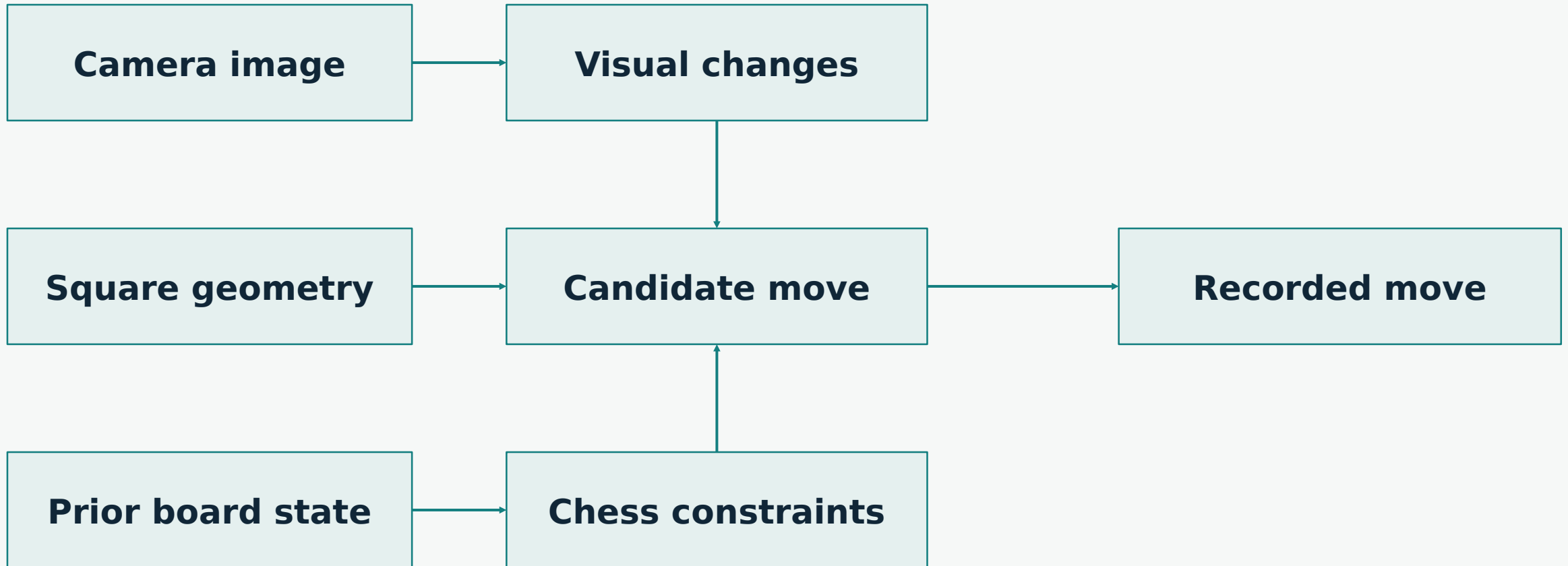
Rigoberto Cortez

Electrical Engineering
University of California San Diego

Embedded systems, computer vision, and physical prototypes

Chess move tracking camera

SPIS 2026 · Working team project



Conceptual overview based on the project description

Tracked line-following robot

PLTW 2024 · Team hardware prototype

Hardware

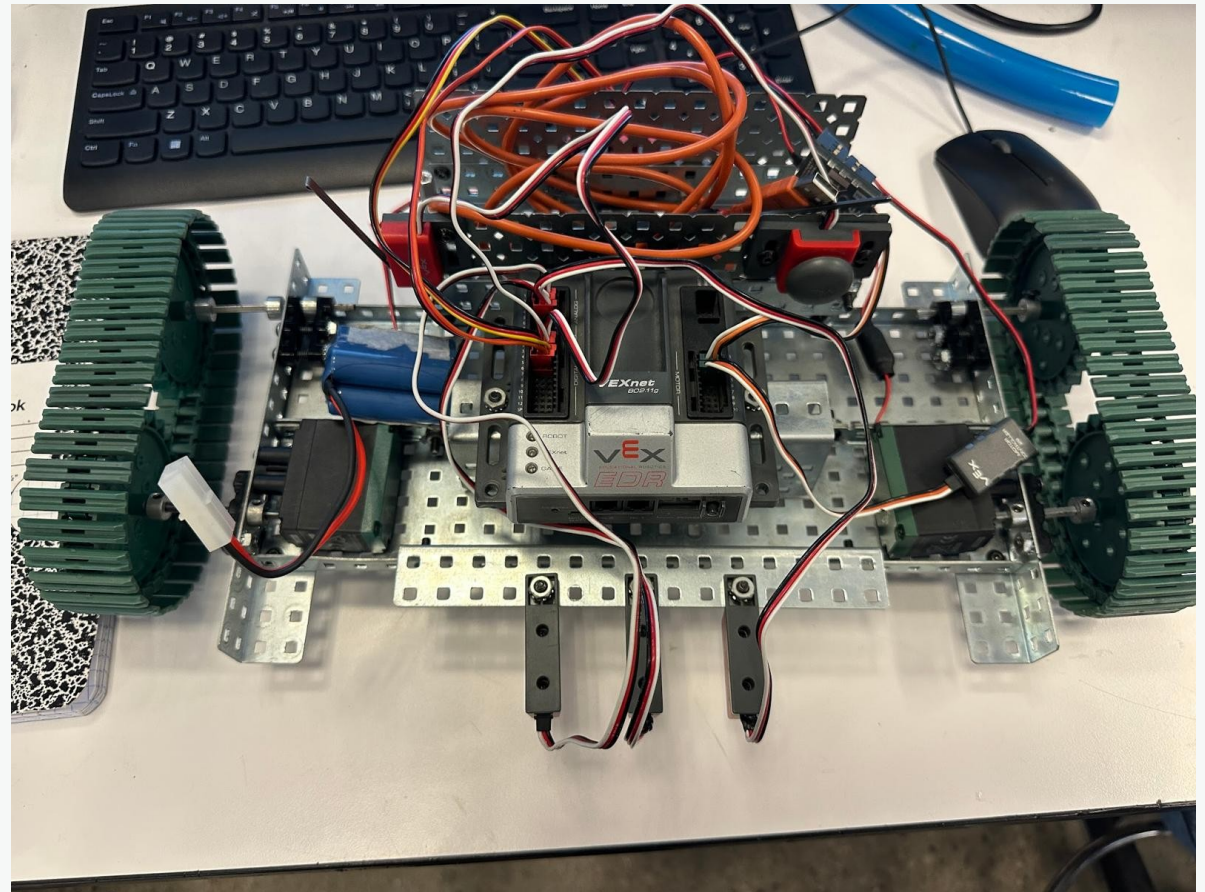
VEX Cortex, two motors, tracks, three line trackers, and ultrasonic sensing.

Integration

Built the chassis and integrated sensors, switching, and wiring. RobotC control work remained incomplete.

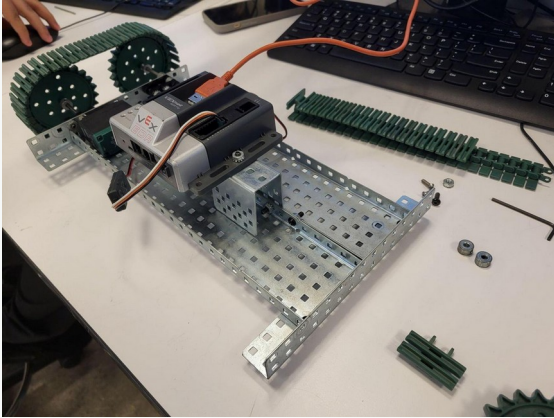
Lesson

Start software trials before the mechanical build is finished.

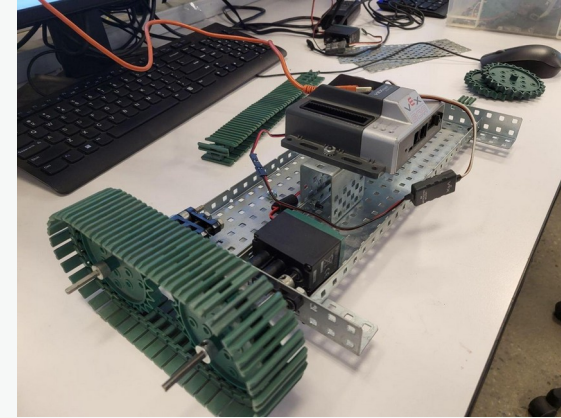


Robot build progression

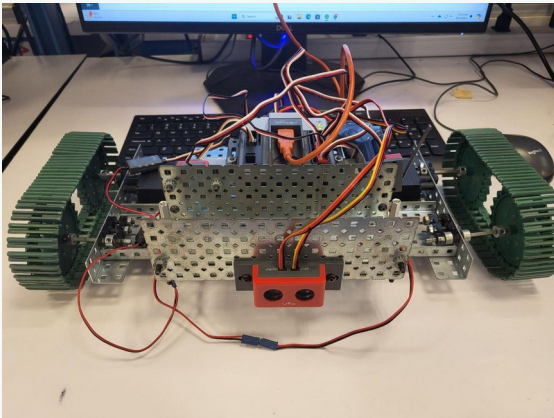
Original build photographs and notebook sketch



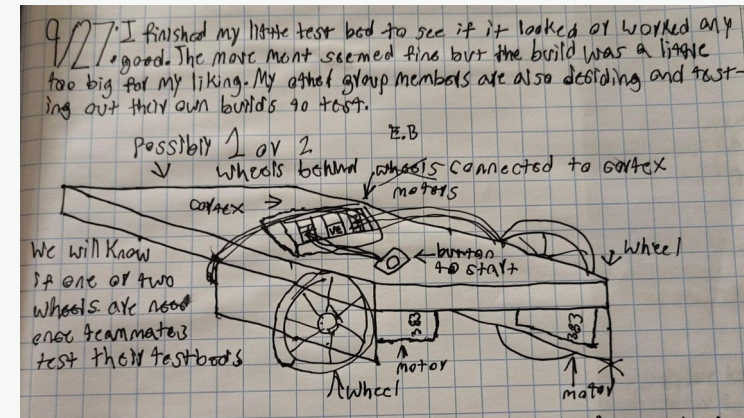
Chassis and drivetrain



Assembly stage



Sensor-facing view



Early concept sketch

Environment-adaptive speaker

ECE Makerspace 2026 · Individual design proposal

Enclosure concept

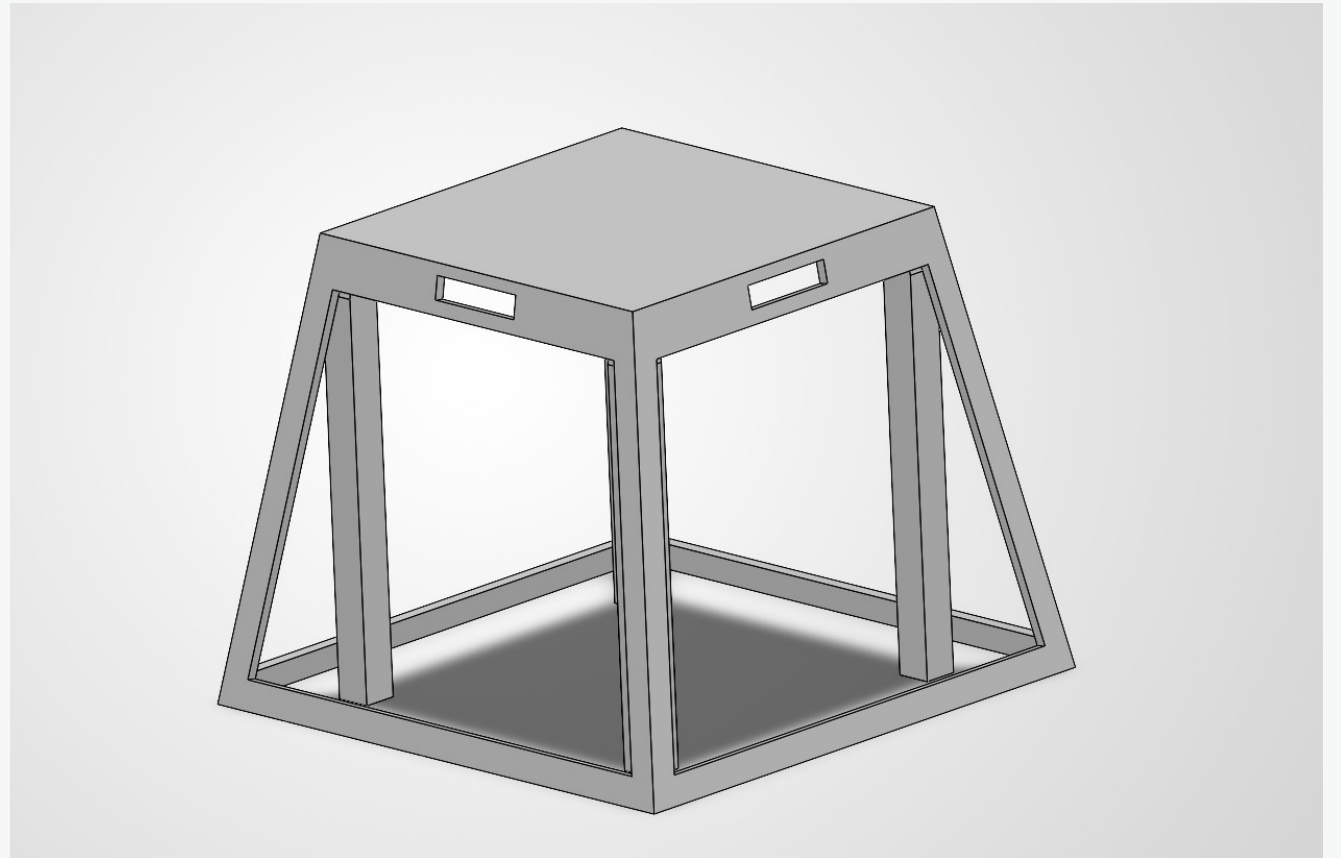
A 6×6 in base and 4.5 in height. Four posts support the truncated pyramid frame.

Adaptive behavior

Each face would sense nearby obstructions and reduce its corresponding driver output.

Completed work

Frame CAD, bottom fastener locations, and a construction plan.



Speaker architecture and testing

Four independent outputs remain a design requirement

Audio control

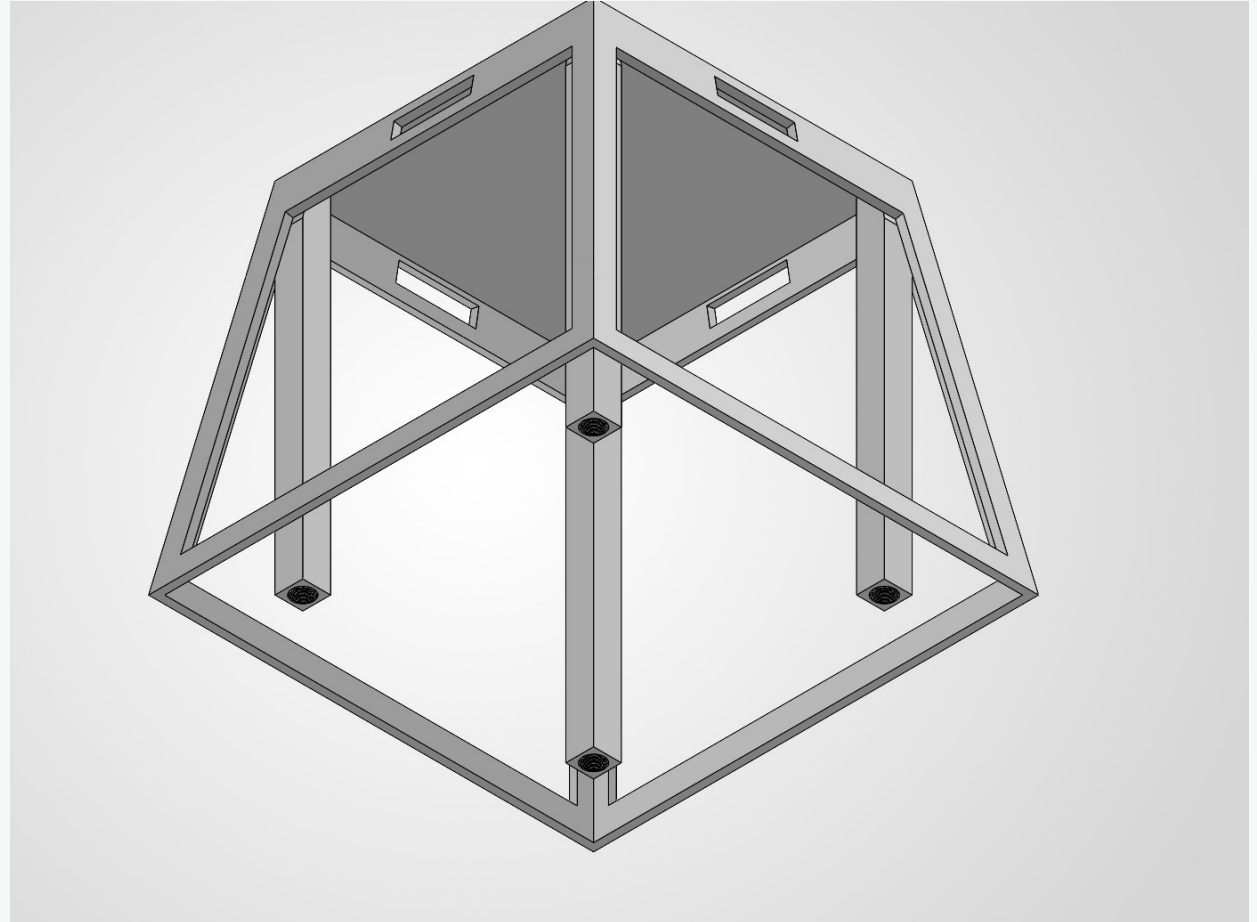
The listed stereo amplifier needs revision or added channel control for four independent outputs.

Test plan

Compare adaptive mode on and off near walls and corners. Measure power and audible effects.

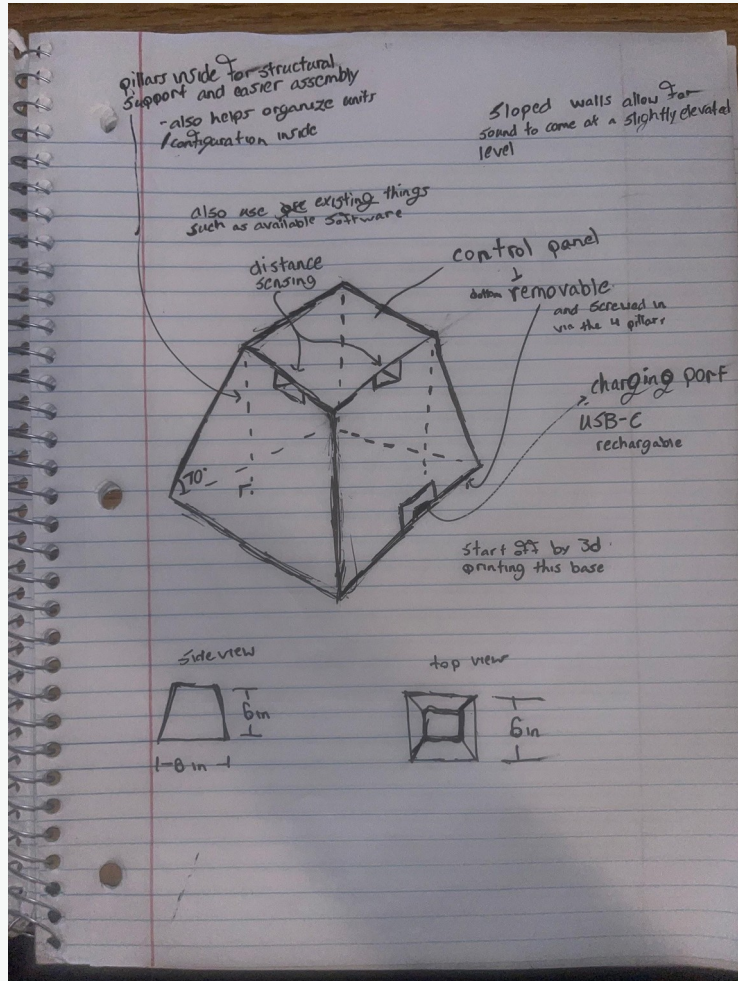
Design questions

Sensor thresholds, driver interaction, and internal volume still need testing.



Speaker concept sketch

Original notebook drawing



Structure

Four internal posts support the enclosure and provide mounting locations.

Access and controls

Top controls and bottom service access appear in the original sketch.

Early concept sketch. Final proposal dimensions appear on the enclosure slide.

Minecraft chess set

PLTW 2024 · Individual parts in a team assembly

My parts

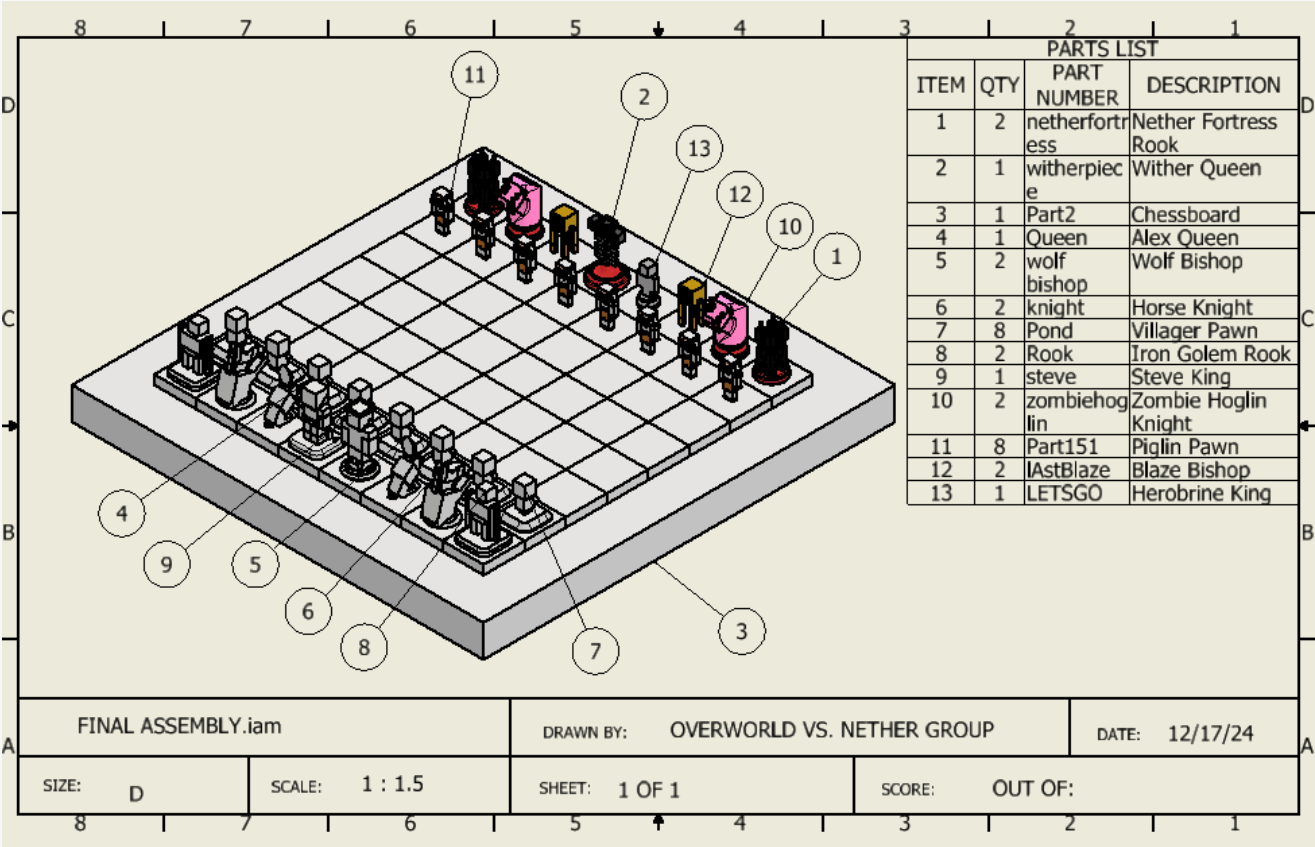
Knight, Steve piece, and wolf bishop. Drawing title blocks identify my contribution.

Documentation

Dimensioned views and neutral STEP geometry preserve the designs.

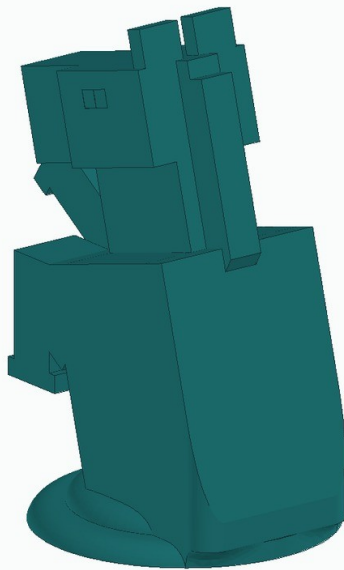
Next iteration

Check manufacturability and base consistency with a physical test print.

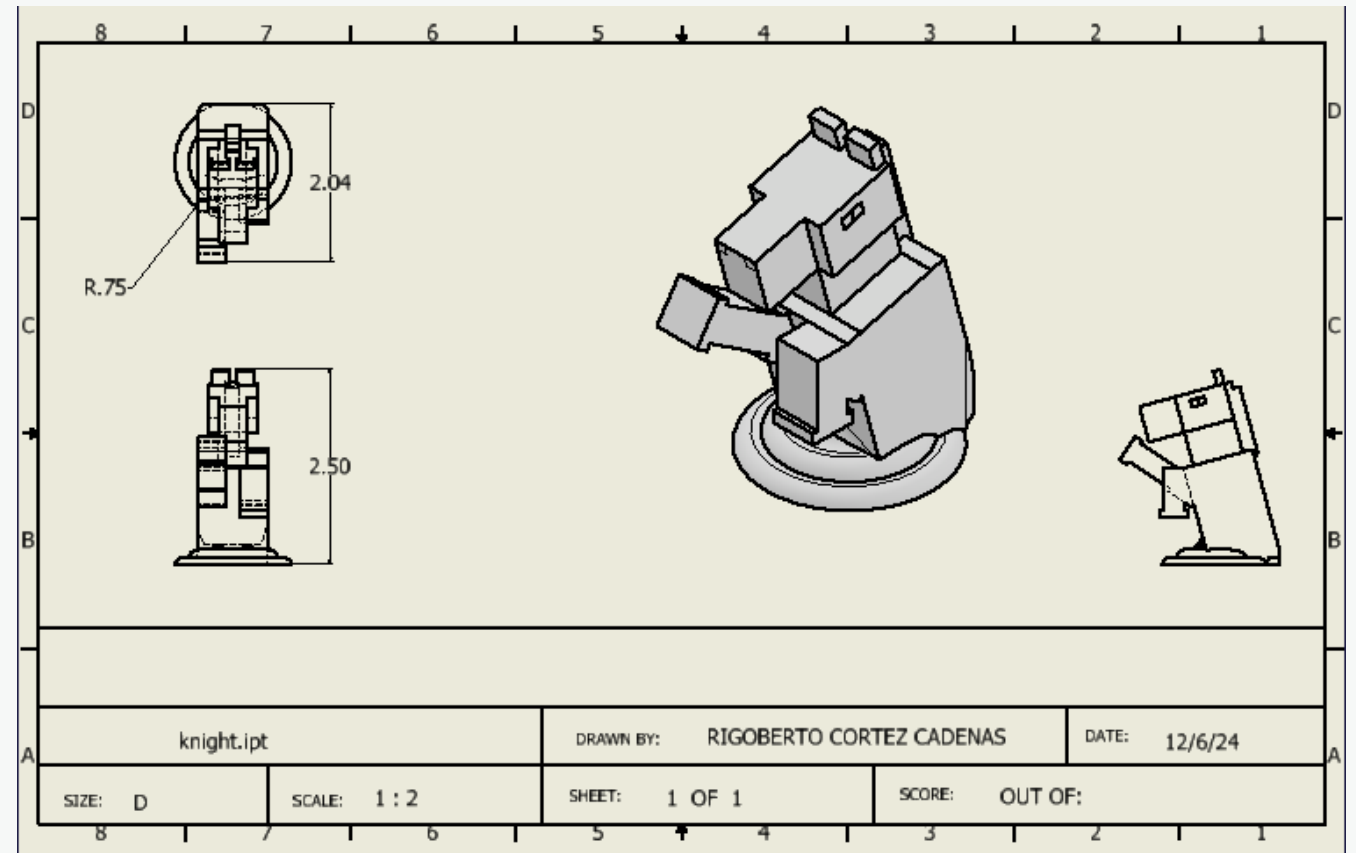


Knight

My Minecraft chess design



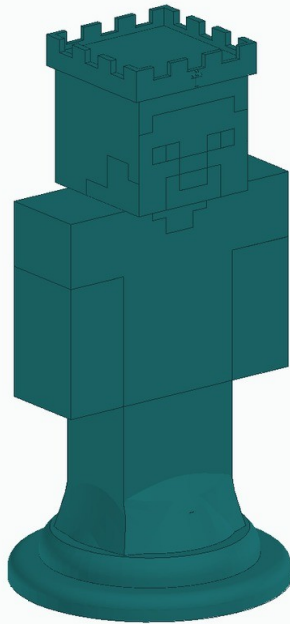
CAD geometry



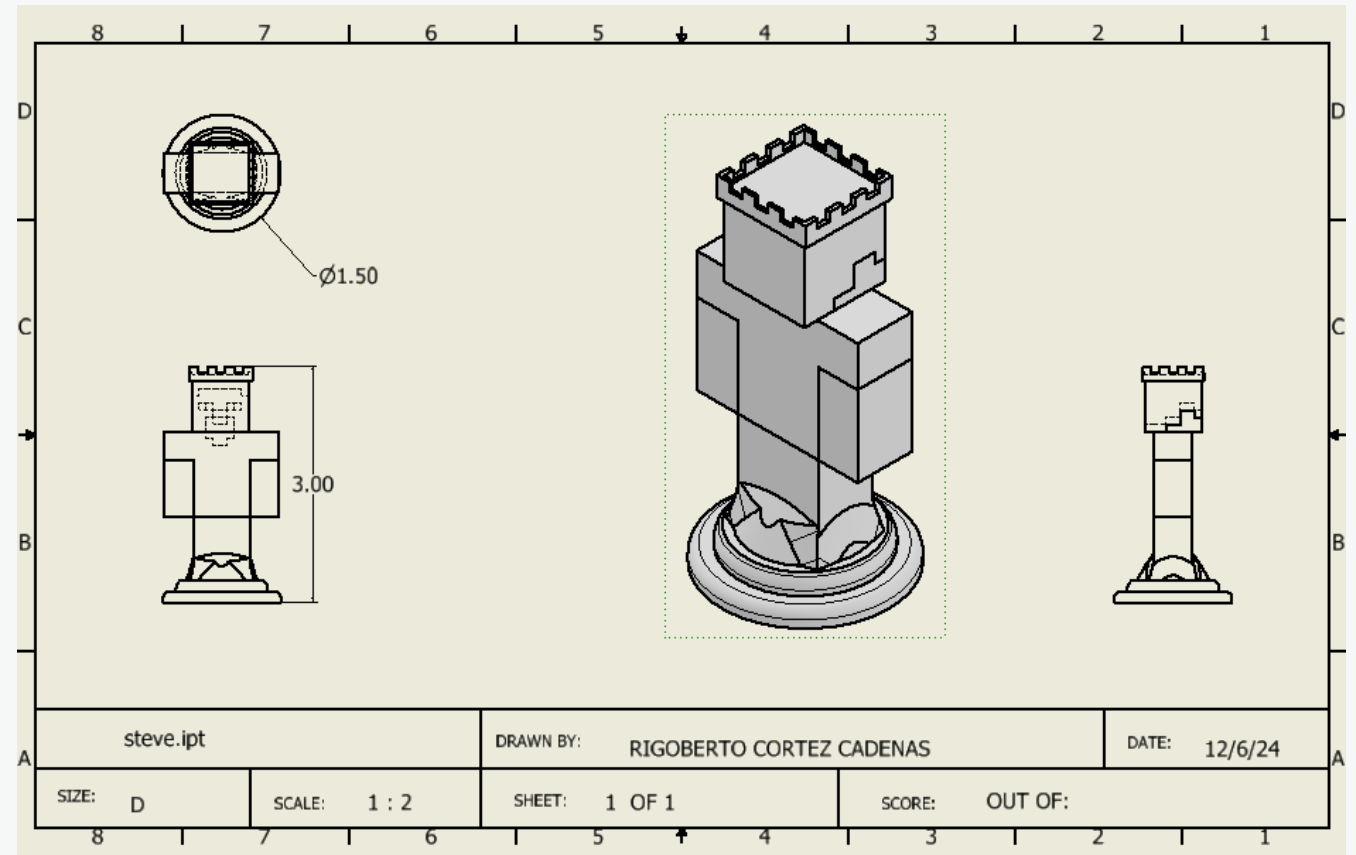
Original engineering drawing

Steve piece

My Minecraft chess design



CAD geometry



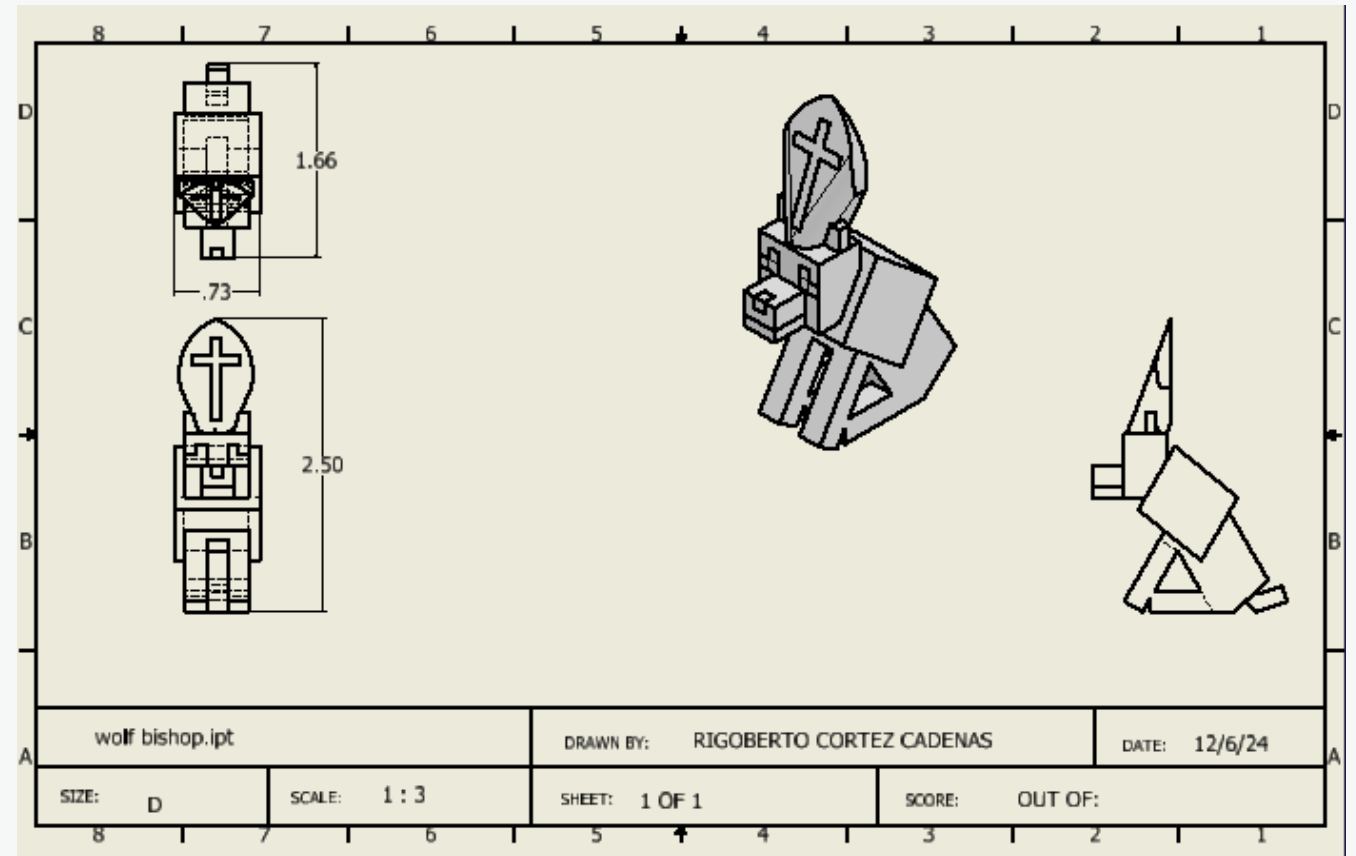
Original engineering drawing

Wolf bishop

My Minecraft chess design



CAD geometry



Original engineering drawing

Star container and CAM studies

PLTW 2025 · CAD models and toolpath simulations

Part geometry

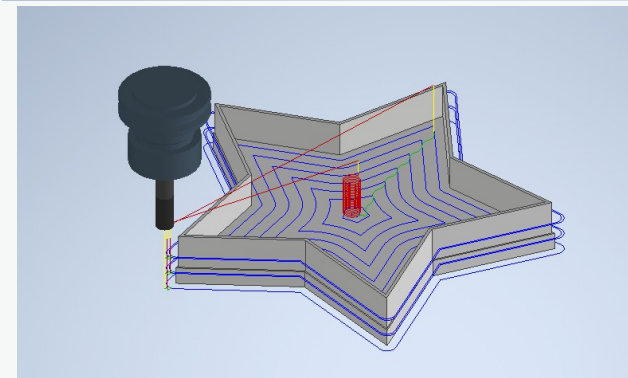
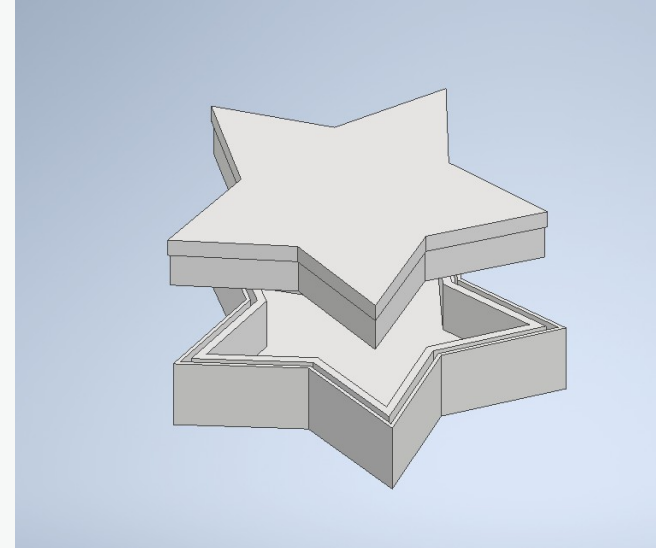
A star-shaped base and fitted lid.
Sketches examine the fit between
the opening and the lid.

CAM work

2D pockets and contours, with
related studies in facing, slotting,
drilling, and surface machining.

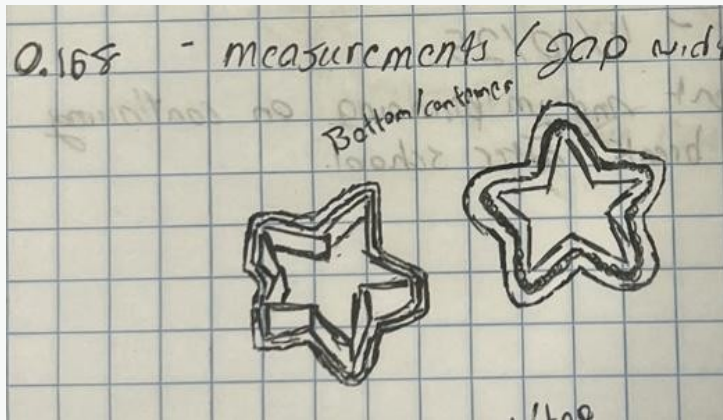
Lesson

Plan tool access while modeling.
Reduce unnecessary tool travel.

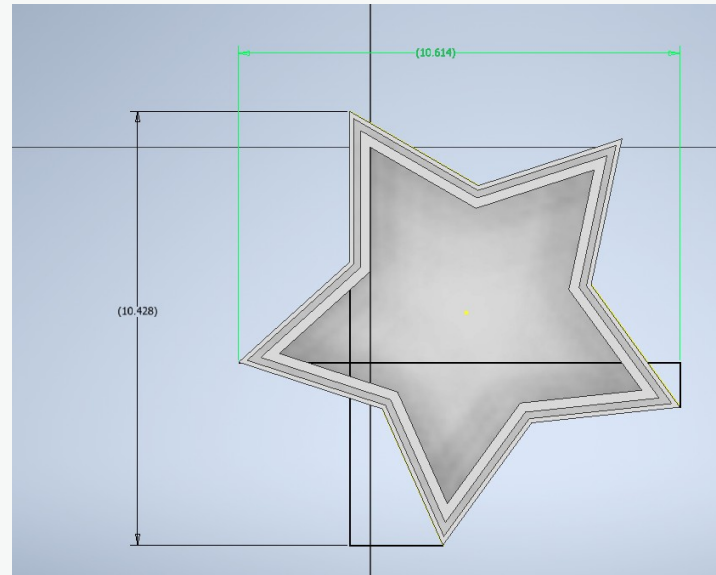


Container sketches and CAM evidence

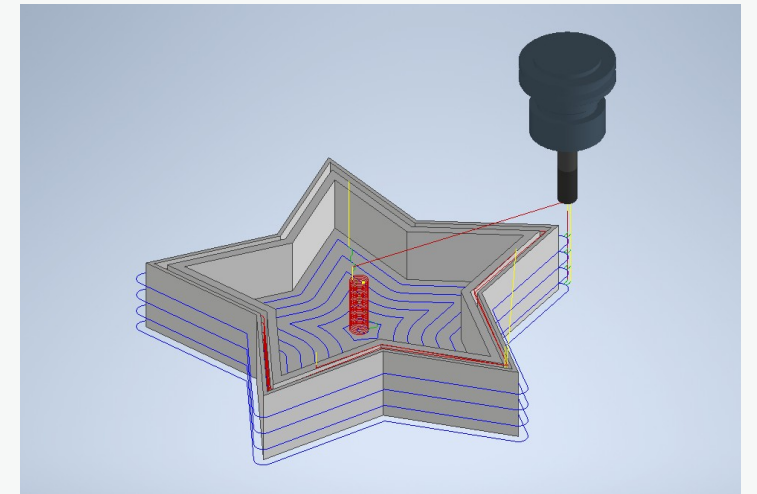
Original design record and machining simulations



Lid fit sketch



Dimensioned top view



Lid toolpath

Bridge build and design reflection

PLTW 2024 · Individual physical build

Constraints

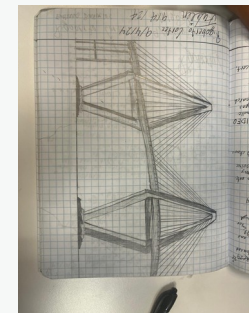
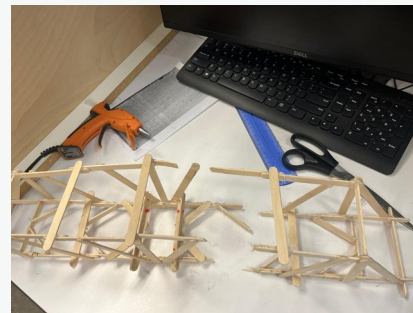
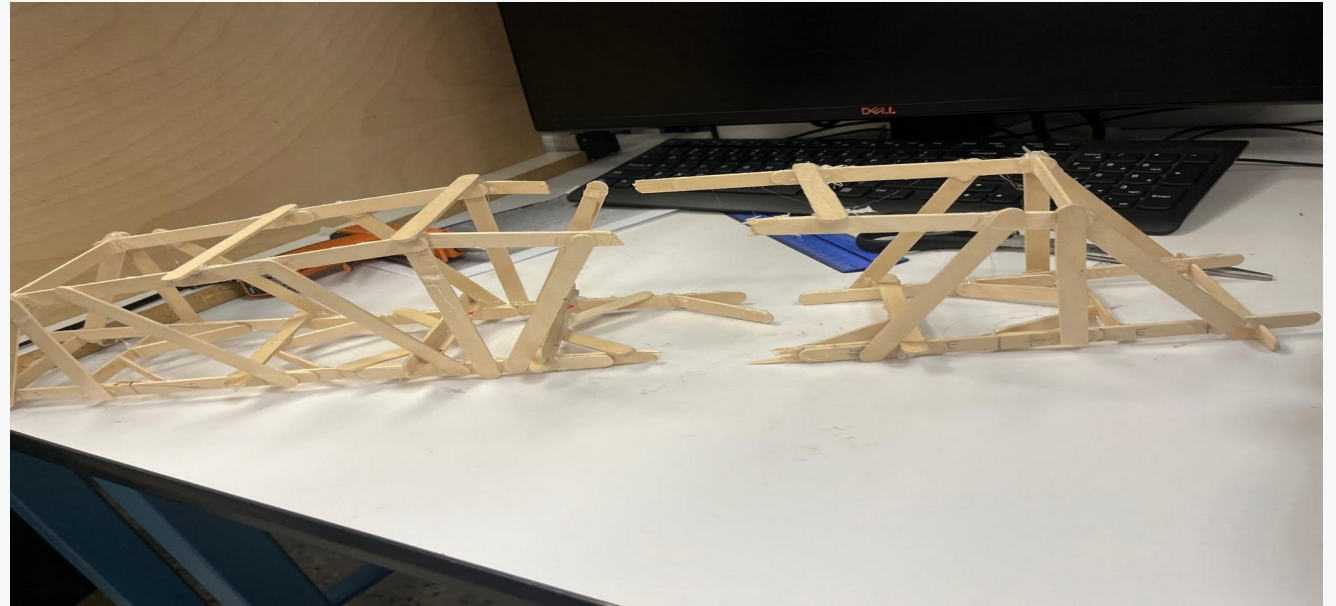
At most 60 popsicle sticks and a required 24-inch span.

Outcome

The bridge carried a load but fell about four inches short of the required span.

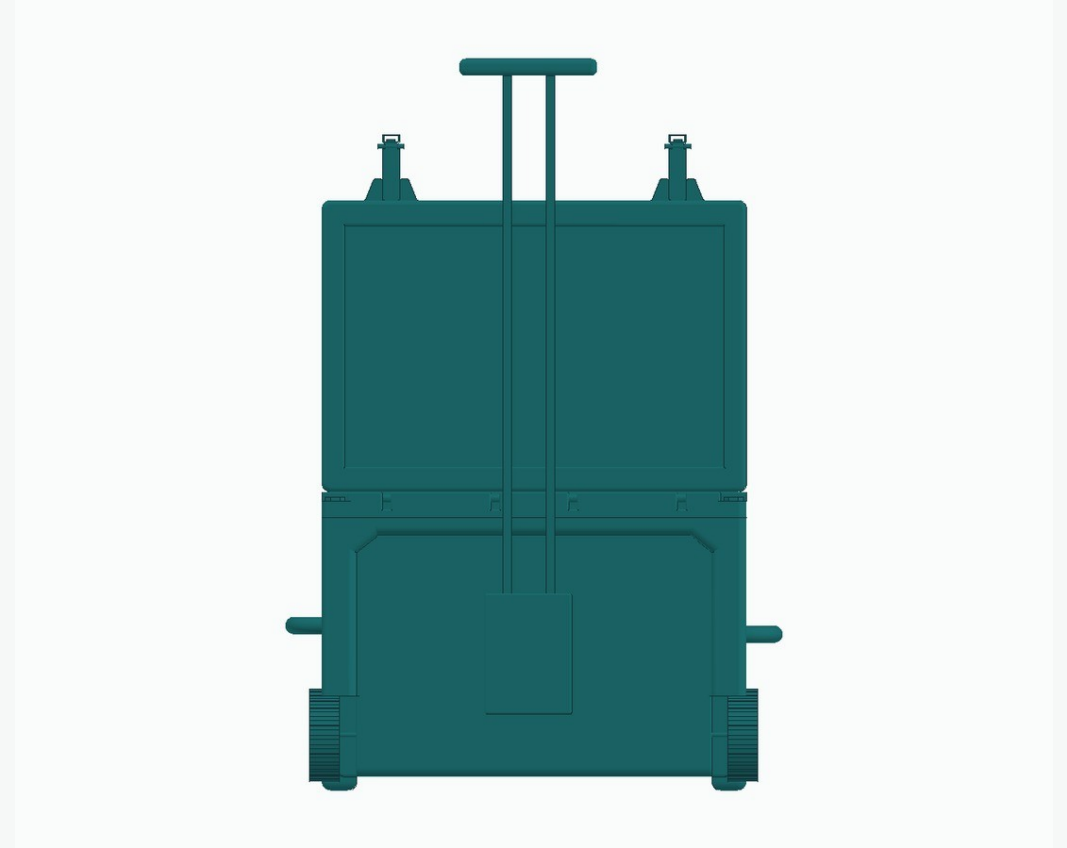
Lesson

Verify dimensions before bracing. A stronger structure still has to meet its original requirements.



Chest cooler team CAD archive

Shared assembly context



Team assembly geometry. My specific contribution still needs documentation.

What I want to build next

Embedded systems that sense the physical world and respond reliably.

I'm developing my C and embedded programming skills.
I'm looking for robotics, hardware, and research teams where I can contribute while continuing to learn.

rcortezcadenas@ucsd.edu
github.com/pur35t/Brief-Projects-